In many sectors in Mathematics, the 'Invariant' objects are quite important.

Def. Let K/F be a finite separable extension, and let L/F be Galois Closure.
Define
$$N_{K/F} : K \to F ; \alpha \mapsto \prod_{\sigma \in \mathrm{Emb}_F(K,L)} \sigma(\alpha)$$
$$Tr_{K/F} : K \to F ; \alpha \mapsto \sum_{\sigma \in \mathrm{Emb}_F(K,L)} \sigma(\alpha).$$
In Particular, if K/F is Galois, then $L=K$, So that $\mathrm{Emb}_F(K/L) = \mathrm{Gal}(K/F)$.

Prop.
1). $N_{K/F}$ and $Tr_{K/F}$ are well-defined. (That is, for any $\alpha \in K$, $N_{K/F}(\alpha), Tr_{K/F}(\alpha) \in F$)
2). The $N_{K/F}$ Preserves multiplication and the $Tr_{K/F}$ Preserves addition.
Proof. Let